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| EXPT NO:6B | <b>DATA VISUALIZATION USING POWER BI (AI IMOACT<br/>ON JOBS ANALYTICS)</b> |
|            |                                                                            |

### AIM:

To load the AI Impact on Jobs 2030 dataset into Power BI, transform and clean the data, and create visualizations to analyze AI-related job patterns and obtain meaningful insights.

### ALGORITHM 1: IMPORTING DATA INTO POWER BI

#### Step 1:

Open Power BI Desktop.

#### Step 2:

Select:

**Home → Get Data**

#### Step 3:

Choose **Text/CSV**.

#### Step 4:

Browse and select the **AI Impact on Jobs 2030 CSV dataset**.

#### Step 5:

Click **Load**.

#### Step 6:

Verify that the dataset has been imported successfully.

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### ALGORITHM 2: DATA TRANSFORMATION USING POWER QUERY

#### Step 1:

Select **Transform Data**.

#### Step 2:

Open Power Query Editor.

#### Step 3:

Remove duplicate records from the dataset.

**Step 4:**

Handle null or missing values in relevant categorical and numerical columns.

**Step 5:**

Rename columns if required and ensure proper formatting of text values such as **Job Title, Industry, Education Level, Automation Level, Company Size, AI Tool Usage, and Hiring Trend**.

**Step 6:**

Check and set appropriate data types:

- Employee ID → Whole Number
- Years Experience → Whole Number
- Future Demand Score → Decimal Number
- Remote Work Possibility → Decimal Number
- Average Salary USD → Whole Number
- AI Replacement Risk → Decimal Number
- Job Growth 2030 → Decimal Number
- Work Hours Per Week → Whole Number
- Performance Score → Decimal Number
- Job Satisfaction → Decimal Number

**Step 7:**

Apply the changes and select **Close & Apply**.

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## ALGORITHM 3: ADVANCED TRANSFORMATIONS

**Step 1:**

Select the **Years Experience** column and verify that its data type is set to Whole Number.

**Step 2:**

Select the **AI Replacement Risk** column and ensure that its data type is Decimal Number.

**Step 3:**

Select the **Future Demand Score** column and ensure that its data type is Decimal Number.

**Step 4:**

Select the **Average Salary USD** column and ensure that its data type is Whole Number.

**Step 5:**

Handle missing values by replacing them with suitable values or removing records containing invalid values.

**Step 6:**

Check categorical columns such as **Industry, Education Level, Automation Level, AI Tool Usage, Upskilling Needed, and Hiring Trend** for inconsistent or blank entries.

**Step 7:**

Filter or remove invalid values from numerical columns such as **Years Experience, Future**

**Demand Score, AI Replacement Risk, Job Growth 2030, Performance Score, and Job Satisfaction.**

**Step 8:**

Click **Close & Apply** to load the cleaned dataset into the Power BI model.

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## **ALGORITHM 4: CREATING CALCULATED COLUMN (AI IMPACT CATEGORY)**

**Step 1:**

Navigate to **Report View** or **Table View**.

**Step 2:**

Right-click the AI Impact on Jobs dataset table and select **New Column**.

**Step 3:**

Create an **AI Impact Category** calculated column to categorize jobs according to AI replacement risk.

Example DAX formula:

```
AI Impact Category =  
IF(  
    [AI_Replacement_Risk] < 0.33,  
    "Low",  
    IF(  
        [AI_Replacement_Risk] < 0.66,  
        "Medium",  
        "High"  
    )  
)
```

**Step 4:**

Press **Enter** to validate and compute the values.

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## **ALGORITHM 5: DASHBOARD VISUALIZATION & REPORTING**

**Step 1:**

Select a **Column Chart** and map **Industry** to the X-axis and **Count of Job Title** to the Y-axis to analyze the distribution of jobs across industries.

**Step 2:**

Select a **Bar Chart** and map **Automation Level** to the X-axis and **Average AI Replacement Risk** to the Y-axis to analyze the relationship between automation and AI impact.

**Step 3:**

Select a **Donut Chart** to show the distribution of jobs based on **AI Impact Category**.

**Step 4:**

Create a **Scatter Chart** using **Years Experience** and **AI Replacement Risk** to analyze their relationship.

**Step 5:**

Create another visualization using **Future Demand Score** and **AI Replacement Risk** to identify patterns related to future job demand.

**Step 6:**

Add **Slicer** visuals for:

- Industry
- Country
- Education Level
- Automation Level
- AI Tool Usage
- AI Impact Category

to enable interactive filtering.

**Step 7:**

Add suitable titles, labels, legends, and data values to the visualizations.

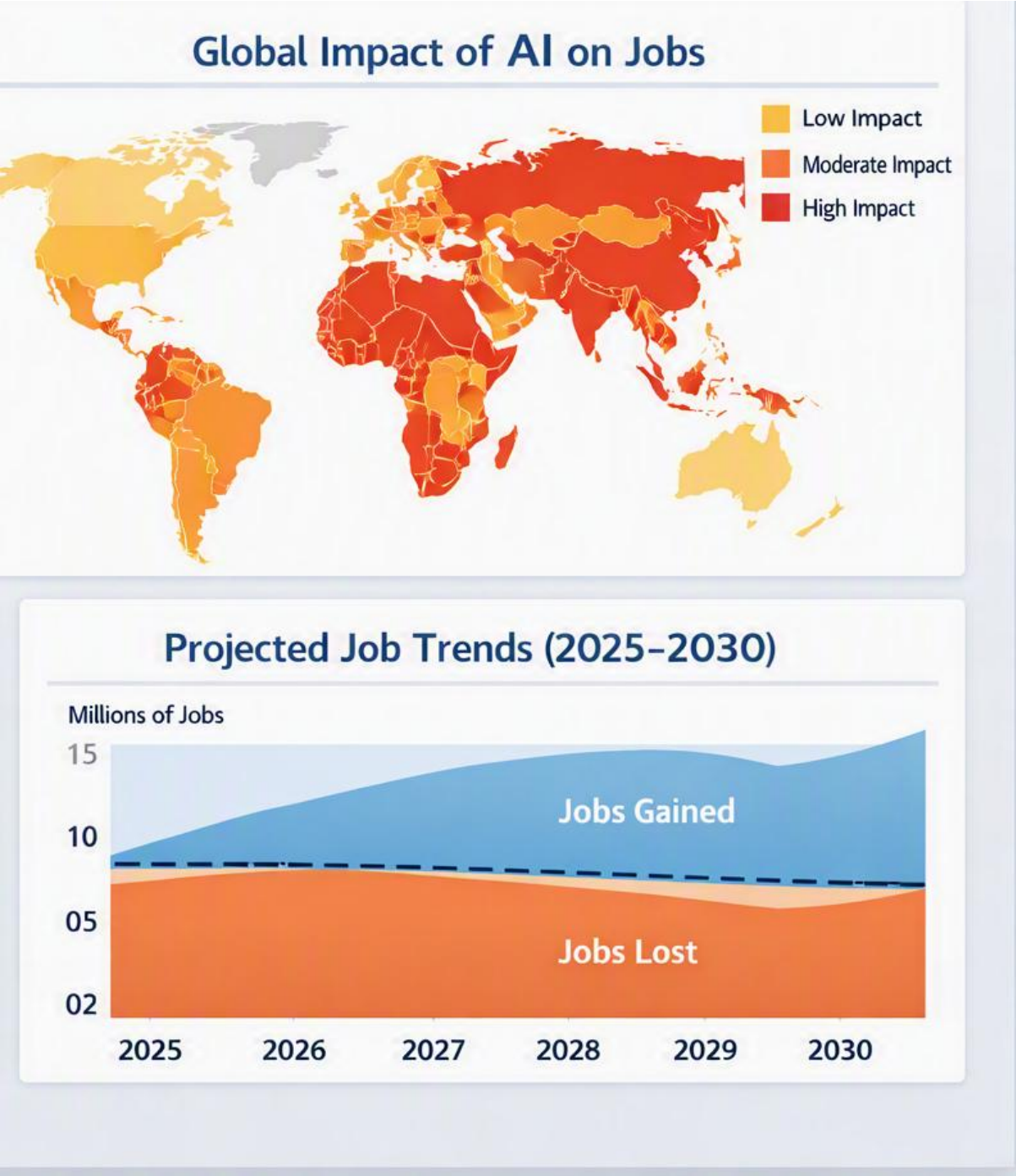
**Step 8:**

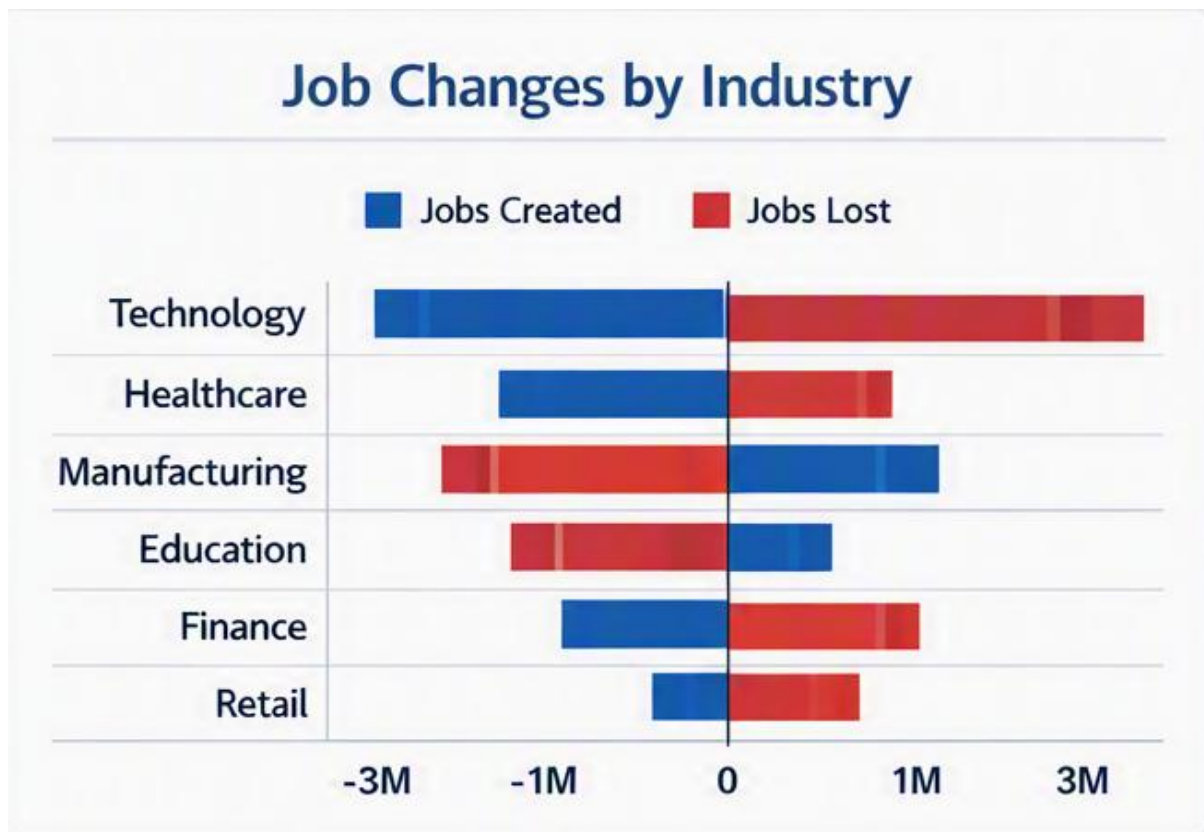
Analyze the dashboard to identify important patterns related to **AI replacement risk, automation, job growth, experience, future demand, salary, and AI tool usage**.

**Step 9:**

Save the Power BI file as a **.pbix** file.

OUTPUT:





### RESULT:

Thus, the Data Visualization of the AI Impact on Jobs 2030 dataset using Python libraries such as Pandas, Matplotlib, and Seaborn was implemented successfully, and various charts were generated to identify important patterns related to AI replacement risk, AI tool usage, salary, and job impact.